1 Any ASE calculator

Quantum chemistry
ORCA, PySCF, ...

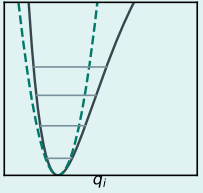
Semi-empirical
GFN2-xTB

Active learning
FALCON

Quantum computing
ADAPT-VQE

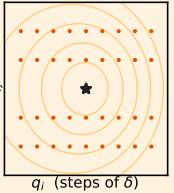
energies, forces, dipoles

2 Harmonic analysis



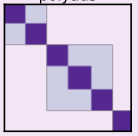
Hessian from forces
symmetrised
Eckart projection
linear molecules,
constrained atoms
→ ω_i, normal modes

3 Quartic force field



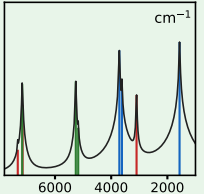
steps in q, $\delta = 0.2$
ϕ_{ijk}, ϕ_{iijj} from E or F
μ_i, μ_{ij} on request
serial or Shephex/SLURM
dry run, restart

4 VPT2



X_{ij} incl. Coriolis
Fermi resonance test
GVPT2 (default)
deperturbed, DCPT2,
standard VPT2
intensity borrowing

5 Spectrum



fundamentals
overtones
combination bands
IR intensities
mid- and near-IR

Numerical noise

Energy errors σ (SCF convergence, ML prediction, VQE optimizer) enter the fourth derivatives as σ/δ^4.

Controlled by tight convergence, the step $\delta = 0.2$, and one consistent surface per finite-difference stencil.